

Task 2C

Solve the following set of problems using Python and submit the code file with extension .ipynb.

1. Load the “[Obesity](#)” dataset. Remove unwanted features if required.
2. Select the optimum k value using Silhouette Coefficient and plot the optimum k values.
3. Create clusters using Kmeans and Kmeans++ algorithms with optimal k value found in the previous problem. Report performances using appropriate evaluation metrics. Compare the results.
4. Now repeat clustering using KMeans for 50 times and report the average performance. Again compare the results that you have obtained in Q3 using Kmeans++ and explain the difference.
5. Apply DBSCAN on this same Obesity dataset and find the optimum "eps" and "min_samples" value. Is the number of clusters the same as the cluster found in Q2? Explain the similarity or differences that you have found between two solutions.
6. Load the “[gene expression](#)” dataset. Apply PCA on the genes for generating 3 principal components. Plot the first three components of the PCA.
7. Continue from question 6, what is the variance (%) covered by the first three components? Explain how this percentage of variance has been computed?
8. Continue from question 6, apply KMeans on the original features of the gene dataset and the first three components returned by PCA. Compare the results using the given labels.

Assessment feedback

The results with comments will be released within 5 business days from the due date.

Referencing

You must correctly use the IEEE method in this assessment. See the [referencing guide](#).

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